

SculptSync

AI exercise-form feedback — pose scoring, LLM coaching evaluation & IoT telemetry

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Project SculptSync (team project) · **Period** May–Oct 2025 · pose-estimation core by the SculptSync team

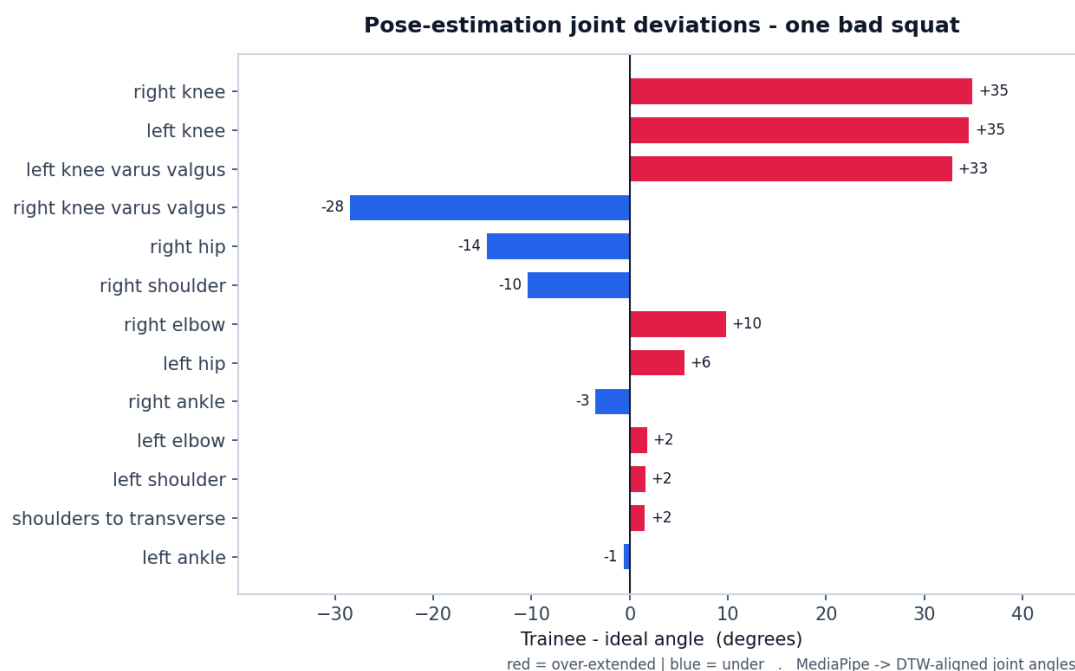
1 What SculptSync does

SculptSync is an AI personal-trainer that scores exercise form and returns targeted, per-rep feedback. A trainee's workout video is compared against ideal reference form: the system generates a form-accuracy score for each repetition and natural-language coaching on exactly which joints to fix — turning raw pose data into advice a beginner can act on.

2 Pose-analysis pipeline

Form is measured by comparing the trainee's movement to an ideal sequence, joint by joint:

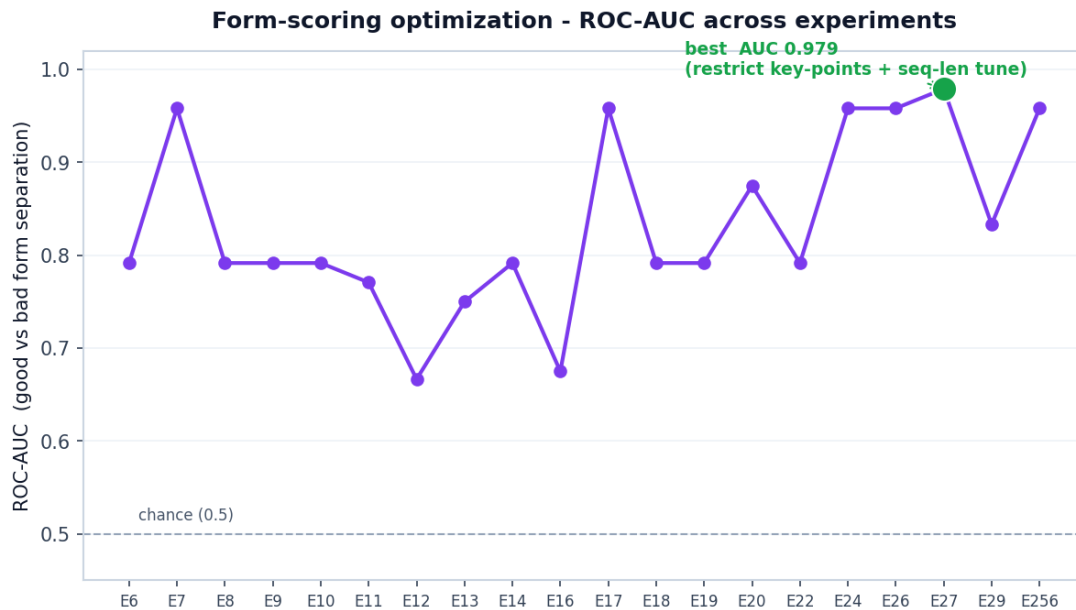
- **MediaPipe** extracts 3D pose landmarks from video frames (sampling every 5 frames).
- **Pose Preprocessor** normalizes for body size, camera position and perspective — filters face/hands, centers on the spine, scales by height, normalizes to $[-1, 1]$.
- **Cosine + Dynamic Time Warping (DTW)** aligns trainee and ideal sequences even at different tempos, with per-landmark distance vectors to locate problem segments.
- **Pose Angle Calculator** derives 3D joint angles — flexion/extension, knee varus/valgus, and shoulder-to-transverse alignment.



Per-joint deviation from ideal for one bad squat — both knees over-extended $\sim 35^\circ$ and valgus collapse flagged automatically. This deviation vector is the signal everything downstream reasons over.

3 Form scoring & optimization

The pipeline condenses each rep into a form-anomaly score (lower = closer to ideal). Quality is measured as ROC-AUC over labelled good- and bad-form clips. I ran a sweep of configurations — restricting key-points to exclude face/hands, key-point weighting, angle normalization, MediaPipe model/confidence tuning and sequence-length tuning — driving ROC-AUC from **0.79** (baseline) to **0.979** (best).



ROC-AUC across optimization experiments; the best run (restricting key-points + sequence-length tuning) reaches 0.979.



At the best configuration, good- and bad-form clips separate cleanly around the decision threshold.

4 LLM feedback evaluation

The joint-angle deviations are handed to an LLM to write the actual coaching. To pick reliable model/prompt pairings instead of trusting demo impressions, I built a benchmark: **7 prompt styles** run across **Gemini 2.0 Flash**, **LLaMA 3** and **Qwen 3**, with Python scoring scripts comparing outputs on

consistency, specificity, safety and actionability — summarised in comparison tables.

LLM feedback evaluation - 7 prompt styles benchmarked

each prompt run across Gemini 2.0 Flash . LLaMA 3 . Qwen 3 -> scored on consistency . specificity . safety . actionability

Standard / Baseline	specific deviations, 2-3 sentences
Clinical (PT-style)	per-joint issue + 3 corrective cues
Trainee-Friendly	plain-English, encouraging tone
Scoring (Quantitative)	rate form /10 + 2 ups / 2 downs
Debugging (Technical)	strength vs mobility vs coordination
Comparative (Multi-video)	which rep is worse & injury risk
Open-Ended (Creative)	vivid in-session coaching description

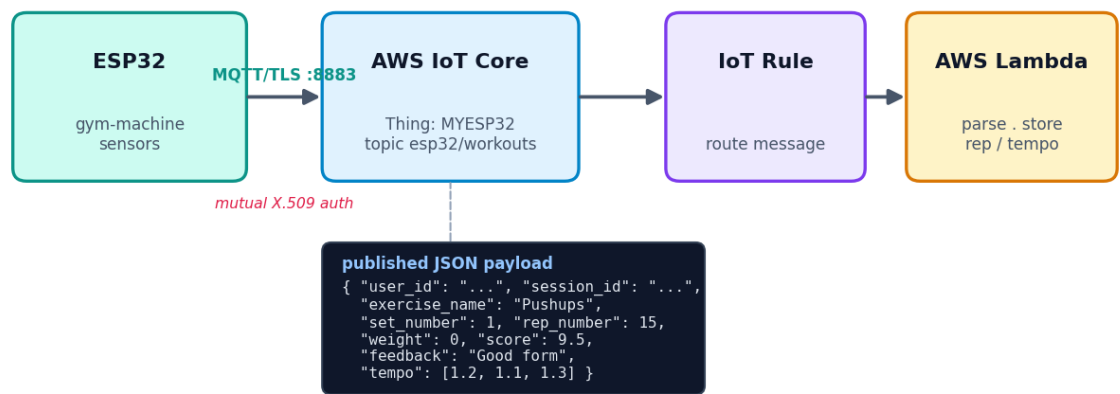
The seven prompt styles benchmarked, from terse PT-style cues to friendly beginner coaching.

5 IoT telemetry pipeline

To ground the feedback in real workout signals, an ESP32 on the gym machine streams telemetry to the cloud. It publishes workout data (reps, sets, tempo, score, feedback) to **AWS IoT Core** over TLS-secured MQTT with mutual X.509 authentication; an IoT Rule routes each message to an **AWS Lambda** that parses and stores it — serverless, cloud-backed logging for repetition-count experiments.

ESP32 -> AWS IoT Core telemetry pipeline

workout telemetry every 10 s . TLS-secured MQTT . serverless ingestion



ESP32 → AWS IoT Core → IoT Rule → Lambda. Device keys and credentials are provisioned on-device and are not shown here.

Team project — the pose-estimation core was built by the SculptSync team. This report covers the LLM feedback evaluation and the IoT telemetry pipeline, contributed by Somesh Agrawal (Part-time ML Engineer, May–Oct 2025).